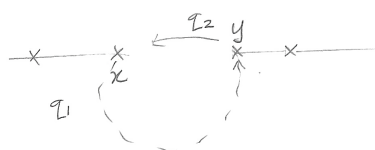


1 Back to basics: Rats and Mice in GPR

Stella worked out that my scheme for multiple steps of the state-dependent approximation in DA does not satisfy detailed balance. Actually, it's pretty easy to show that it fails, by being a little careful ...

17/7/26

①



proposal q_1 at x targets π_1 , then MH step targets π
 q_2 at y targets π_2

if q_1 targets π_1
 q_2 targets π_1 } does MH acceptance prob. simplify

propose $q_1^n(y|x)$ accept w.p. $1 \wedge \frac{q_1^n(x|y) \pi(y)}{q_1^n(y|x) \pi(x)} = 1 \wedge \frac{\pi(x)}{\pi_1(y)}$

q_1 in detailed balance with π_1 , $q_1(x|y) \pi_1(y) = q_1(y|x) \pi_1(x)$

q_2 " " " " π_2 $q_2(x|y) \pi_2(y) = q_2(y|x) \pi_2(x)$

also $\pi_1(x) = \pi(x)$
 $\pi_2(y) = \pi(y)$

$$1 \wedge \frac{\pi_1(x) \pi(y)}{\pi_1(y) \pi(x)} = 1 \wedge \frac{\pi(y)}{\pi_1(y)}$$

call $q_1^n(y|x) (1 \wedge \frac{\pi(y)}{\pi_1(y)}) = q_1^*(y|x)$ (effective proposal) is in detailed balance with π .

$$q_1^*(y|x) \pi(x) = q_1^n(y|x) \pi(x) \wedge \frac{q_1^n(x|y) \pi(y)}{\pi_1(y)}$$

$$(q_1^n(y|x) \wedge \frac{q_1^n(x|y) \pi(y)}{\pi_1(y)}) \pi(x) = q_1^n(y|x) \pi(x) \wedge q_1^n(x|y) \pi(y) \quad \begin{matrix} \text{symmetry} \\ \Rightarrow \text{result} \\ \text{so detailed balance holds.} \end{matrix}$$

$$\text{ie } q_1^*(y|x) \pi(x) = q_1^*(x|y) \pi(y)$$

$$\text{why } q_2^*(y|x) \pi(x) = q_2^*(x|y) \pi(y)$$

Now propose with $q_1^*(y|x)$ to $\pi(x)$. accept/reject with $1 \wedge \frac{q_1^*(y|x) \pi(x)}{q_2^*(x|y) \pi(y)}$

$$= 1 \wedge \frac{q_1^*(x|y) \pi(y)}{q_2^*(x|y) \pi(y)} \quad \text{not } = 1!$$

Instead, I will give up on the MH correction, and see if ergodicity can be achieved more directly.

1.1 Ergodicity

I use the definition of *ergodicity* in its asymptotic sense, because that's the property we want. Let P be a Markov transition kernel with invariant distribution π . We say that P is *ergodic for* π if, for every initial distribution μ ,

$$\mu P^n \longrightarrow \pi,$$

or, more weakly, if the corresponding Markov chain satisfies the ergodic theorem

$$\frac{1}{N} \sum_{k=0}^{N-1} f(X_k) \longrightarrow \int f d\pi$$

almost surely for every integrable function f .

The first property, convergence in distribution, is sufficient to establish the second, namely almost sure convergence of Monte Carlo estimates. For Monte Carlo, it is the latter property that is fundamental, since it guarantees the consistency of estimates of expectations with respect to π .

In much of the finite-state Markov chain literature, the term *ergodic* is instead defined to mean *irreducible and aperiodic*. While this terminology is convenient, these are in fact structural assumptions on the transition graph, from which the asymptotic convergence theorem is deduced. The theorem goes back to Perron and Frobenius. Standard modern references are:

- **Seneta, E. (2006). Non-negative Matrices and Markov Chains. Springer.** (Revised printing of the classic text.) This develops Perron-Frobenius theory and proves that a finite irreducible, aperiodic stochastic matrix has a unique stationary distribution and that P^n converges to the rank-one limit.
- **Levin, D. A., Peres, Y., & Wilmer, E. L. (2009). Markov Chains and Mixing Times. American Mathematical Society.** Chapter 1 proves that irreducibility and aperiodicity imply convergence to the unique stationary distribution in finite state spaces. This is probably the modern reference most readers would recognize.
- **Norris, J. R. (1997). Markov Chains. Cambridge University Press.** See the early chapters on finite-state chains. A very clean mathematical treatment.
- **Kemeny, J. G. & Snell, J. L. (1976). Finite Markov Chains. Springer.** The classic older reference, available online.

For finite state spaces these graph-theoretic conditions characterize ergodicity in the above asymptotic sense, but conceptually they are quite different; the definition I use defines the desired limiting behaviour, while the other specifies sufficient (indeed, equivalent) graph-theoretic conditions used to establish it.

This distinction is important in our work. Our primary concern is the asymptotic behaviour of Markov chains and the correctness of Monte Carlo estimates, rather than the graph-theoretic conditions by which convergence is established. We therefore regard irreducibility and aperiodicity as properties of the underlying transition graph, not as the definition of ergodicity itself.

However, from the graph-theoretic viewpoint, irreducibility and aperiodicity have simple interpretations. A finite-state Markov chain defines a directed graph whose vertices are the states and whose edges correspond to transitions of positive probability. Irreducibility is precisely the statement that this graph is strongly connected: every state can be reached from every other state. Aperiodicity is a property of the cycle structure of this graph. It excludes cyclic decompositions of the graph by requiring that the greatest common divisor of the lengths of all directed cycles through any state is one. Equivalently, the graph is not constrained to return to a state only at multiples of some integer greater than one. In many Markov chain Monte Carlo algorithms, aperiodicity is almost automatic. For example, in the Metropolis–Hastings algorithm, proposed moves are rejected with positive probability, so every state has a positive probability of remaining unchanged. The resulting self-loops immediately destroy periodicity, provided the chain is irreducible. Thus, both irreducibility and aperiodicity are fundamentally graph-theoretic properties rather than probabilistic notions. In practical MCMC algorithms, aperiodicity is usually straightforward to ensure through the presence of self-loops arising from rejected proposals.

For the finite-state examples the we consider first, these graph properties provide convenient sufficient conditions for convergence to the invariant distribution. Nevertheless, our emphasis remains on ergodicity as an asymptotic property of the Markov process rather than on the graph-theoretic assumptions used to establish it. Our interest is less in identifying graph-theoretic hypotheses under which ergodicity holds, and more in constructing transition mechanisms that are ergodic for a prescribed target distribution.

1.2 Distributional convergence is not actually necessary for ergodic in the sense of consistency of Monte Carlo estimates

Take the cyclic permutation R on n states. We want ergodicity wrt the uniform distribution. Clearly,

$$\mu R^n$$

does not converge unless μ is already uniform, so by the first criterion R is not ergodic for the uniform distribution.

However, if f is any function and $X_{k+1} = R(X_k)$, then for every starting state

$$\frac{1}{N} \sum_{k=0}^{N-1} f(X_k) \rightarrow \frac{1}{n} \sum_{i=1}^n f(i)$$

because the trajectory simply cycles through every state exactly once every n steps. So the ergodic averages are perfect, even though the distributions never converge. That's a fundamental distinction. In dynamical systems language, R is uniquely ergodic (on its single periodic orbit) in the sense of Birkhoff averages, but it is not mixing. In finite Markov-chain language, periodicity prevents convergence of the marginals while leaving Cesàro averages untouched.

1.3 Finite state space

What can we say about state-dependent kernels in the simple case of finite number of states?

Consider a $n \times n$ kernel P designed to converge to a target distribution π on n states. For simplicity (and I think wlog) take the target to be the uniform distribution on n states

$$u = \frac{1}{n}(1, \dots, 1)^T = \frac{1}{n}\mathbf{1}.$$

Lemma: P has u as an invariant distribution iff P is doubly stochastic.

Proof: Let P be a row-stochastic matrix (rows sum to 1, i.e., $P\mathbf{1} = \mathbf{1}$). Since u is invariant under P ,

$$u^T P = u^T.$$

Taking transpose

$$P^T u = \frac{1}{n}\mathbf{1} \text{ or } P^T \mathbf{1} = \mathbf{1},$$

i.e. the columns of P sum to 1.

For $n = 3$, all possible such P have the explicit form:

$$P = \begin{pmatrix} a & b & 1-a-b \\ c & d & 1-c-d \\ 1-a-c & 1-b-d & a+b+c+d-1 \end{pmatrix},$$

and the feasible region is the inequalities

$$\begin{aligned} a, b, c, d &\geq 0, \\ a+b &\leq 1, \quad c+d \leq 1, \\ a+c &\leq 1, \quad b+d \leq 1, \\ a+b+c+d &\geq 1. \end{aligned}$$

That's a four-dimensional polytope.

Most of these are not symmetric. For example,

$$P = \begin{pmatrix} 0.6 & 0.4 & 0 \\ 0 & 0.6 & 0.4 \\ 0.4 & 0 & 0.6 \end{pmatrix}$$

is doubly stochastic but not symmetric (and is circulant, non-normal). Its eigenvalues are

$$1, \quad 0.4 + 0.3464i, \quad 0.4 - 0.3464i,$$

with the complex eigenvalues having modulus about 0.529 (less than 1). Hence

$$P^n \rightarrow \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

which means that P is ergodic for u , by convergence in distribution. So symmetry plays no essential role – which I mention because detailed balance is exactly a statement that P is symmetric.

1.4 The Birkhoff polytope

A lovely mathematical object is the Birkhoff polytope of doubly stochastic matrices, though I'm not sure it is actually useful for our purposes.

The set of all $n \times n$ doubly stochastic matrices is a convex polytope

$$\mathcal{B}_n = \{P \geq 0 : P\mathbf{1} = \mathbf{1}, \mathbf{1}^T P = \mathbf{1}^T\}.$$

The dimension is

$$(n-1)^2.$$

Indeed there are n^2 entries, $2n$ linear constraints, but one is redundant, as is the basis. For $n = 3$,

$$\dim \mathcal{B}_3 = 4,$$

as we saw above.

The remarkable Birkhoff-von Neumann theorem is: The extreme points of $\mathcal{B}_n$ are the permutation matrices.

Equivalently: Every doubly stochastic matrix is a convex combination on of permutation matrices.

That is

$$P = \sum_{k=1}^{n!} \alpha_k \Pi_k,$$

where

$$\alpha_k \geq 0, \quad \sum \alpha_k = 1.$$

For $n = 3$, there are six vertices. They are the six permutations of three objects. So every 3×3 doubly stochastic matrix lives inside the convex hull of those six matrices.

Note that $\sum \alpha_k = 1$ is one constraint. At first sight, that's six variables minus one constraint, i.e. five degrees of freedom. But $\mathcal{B}_3$ is only 4-dimensional, because the representation is not unique. There is a linear dependence among the six permutation matrices. If we denote the three even permutations by E_1, E_2, E_3 and the three odd ones by O_1, O_2, O_3 , then

$$E_1 + E_2 + E_3 = O_1 + O_2 + O_3.$$

That's another relation. Consequently every point in $\mathcal{B}_3$ has infinitely many convex decompositions into permutation matrices. So the convex coefficients are not nice coordinates.

1.5 State dependent kernels

When the chain is in state x , apply kernel P_x .

Suppose the state space is $\{1, \dots, n\}$, and each P_i is a stochastic matrix. Then the transition probability from x to y is now

$$Q_{xy} = (P_x)_{xy},$$

i.e. the x -th row of Q is the x -th row of P_x . So the family $\{P_x\}$ defines a new Markov chain with transition matrix

$$Q = \begin{pmatrix} \text{row}_1(P_1) \\ \text{row}_2(P_2) \\ \vdots \\ \text{row}_n(P_n) \end{pmatrix}.$$

That means we are not studying products of kernels at all – we build a new kernel by selecting rows.

Proposition: ($\mathbf{n} = \mathbf{3}$). Every probability vector is the first row of a symmetric doubly stochastic 3×3 matrix. In other words, for $n = 3$,

$$R_1 = \Delta_2$$

the entire probability simplex. So, reversibility imposes no restriction on an individual row.

Proof: Starting with

$$r_1 = (a, b, c),$$

take

$$P = \begin{pmatrix} a & b & c \\ b & c & a \\ c & a & b \end{pmatrix}.$$

The general $n \times n$ case has the same conclusion.

Lemma 1.5.1. *Every probability vector is the first row of a symmetric doubly stochastic matrix.*

Consequently, in a row-wise construction, requiring each underlying kernel to be reversible with respect to the uniform distribution places no restriction on the admissible row itself.

Proof. Let

$$r = (r_0, r_1, \dots, r_{n-1})$$

be an arbitrary probability vector, with

$$r_k \geq 0, \quad \sum_{k=0}^{n-1} r_k = 1.$$

Define the matrix P by (the construction)

$$P_{ij} = r_{(i+j) \bmod n}, \quad i, j = 0, \dots, n-1.$$

Since addition modulo n is commutative,

$$P_{ij} = P_{ji},$$

so P is symmetric.

For fixed i , the map

$$j \mapsto (i+j) \bmod n$$

is a permutation of $\{0, \dots, n-1\}$. Hence every row of P is a permutation of the entries of r , and therefore has sum

$$\sum_{j=0}^{n-1} P_{ij} = \sum_{k=0}^{n-1} r_k = 1.$$

Similarly, for fixed j , the map

$$i \mapsto (i+j) \bmod n$$

is also a permutation, so every column is a permutation of the entries of r , and therefore also sums to one.

Thus P is symmetric and doubly stochastic, with first row equal to r . Consequently every probability vector is the first row of a symmetric doubly stochastic matrix. $\square$

Corollary: By permutation of rows and columns, a probability vector can be any row of a symmetric doubly stochastic matrix.

The construction in the Lemma is not unique, so there may be many such matrices. E.g., for $n = 4$, start with the cyclic Latin square (of indices)

$$\begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 0 \\ 2 & 3 & 0 & 1 \\ 3 & 0 & 1 & 2 \end{pmatrix}.$$

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Now, in rows 2 and 4 (using 1-based indexing), swap the symbols 0 and 2 . The result is

$$\begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 1 \\ 3 & 2 & 1 & 0 \end{pmatrix},$$

which is still a symmetric Latin square.

All that means that detailed balance wrt π of each P_x is virtually no restriction on what a row of P_x can look like. Bugger!

However, each row, in particular the x -th row, of P_x must come from a doubly stochastic matrix (assuming detailed balance wrt π), so that may be some restriction.

We want the resulting kernel Q to target u . The question then is, how do we ensure that Q is doubly stochastic, i.e., columns sum to 1? Presumably we need to scale the rows of Q in some way. How? The problem with taking $n > 1$ steps using a state-dependent approximation is that we do not have access to the elements of P_x , hence Q . Bugger!

2 Basins and mountain passes

I think I might change tack and look at the problem from another view point. My target distribution, like many, can likely be characterised by regions of high probability (low energy/misfit) separated by regions of low probability, perhaps with channels joining the energy basins along which the state may move. My previous concern was with making Markov kernels that transit the channels – using state-dependent approximations within a multi-step delayed acceptance algorithm and basic moves such as birth/death, perturb position, etc. The picture of energy basins and "mountain passes" is something I have seen in computational quantum chemistry. My thought is to treat each basin as a node in a graph of basins, with connections discovered by my approximate multi-step scheme, and then make a meta Markov chain that jumps between basins with weights determined by volume (of probability) in each basin, and some rapid scheme of sampling each basin, e.g. by Gaussian approximation.

I've seen this idea in quantum chemistry – but typically they optimise (move between minima), not sample. See:

- **David J. Wales. 2018. Exploring Energy Landscapes. Annual Review Physical Chemistry. 69:401-425.** <https://doi.org/10.1146/annurev-physchem-050317-021219>
- **Yang, S., Day, G.M. Global analysis of the energy landscapes of molecular crystal structures by applying the threshold algorithm. Commun Chem 5, 86 (2022).** <https://doi.org/10.1038/s42004-022-00705-4>

Markov state models (MSMs) is perhaps the closest mathematical framework. One first partitions state space into metastable basins. One then estimates $P_{ij} = \text{Pr}(\text{next basin} = j \mid \text{current basin} = i)$. The resulting Markov chain approximates the long-time dynamics. MSMs are now standard in protein folding and molecular dynamics – where AI has been used to great effect. See, e.g.,

- **Pearce, P., Woodhouse, F.G., Forrow, A. et al. Learning dynamical information from static protein and sequencing data. Nat Commun 10, 5368 (2019).** <https://doi.org/10.1038/s41467-019-13307-x>

I know that Jun Liu has worked in this area, so might be useful looking up his work.

My suggested scheme is somewhat different to the standard in the chemistry literature, that generally proceeds

1. sample dynamics
2. infer basins
3. estimate transition matrix.

I propose almost the reverse. We use our multi-step approximation machinery to

1. discover neighbouring basins,
2. estimate the probability mass in each basin, (using efficient continuous-variable samplers),
3. then construct a Markov chain on basins (plus moves within basins).

So the graph that shows connections between basins becomes part of the proposal mechanism.

Graph transition weights are determined by volume (of probability) in each basin:

$$\pi_i = \int_{B_i} \pi(x) dx$$

where B_i is basin i . Now the chain lives on

$$\{B_1, \dots, B_m\},$$

rather than on the original state space. Inside each basin one could sample rapidly using some flashy continuous-variable sampler.

Actually, now I see that the graph is irrelevant, it's just the probability mass of each configuration that matters, setting the frequency of visiting via sampling of the discrete distribution over masses.

2.1 No graph, no passes, just probability mass in basins

The notion of “mountain passes” comes from studying the *dynamics* of molecules – how does one molecule deform to give another? But we are only interested in the equilibrium distribution, so it does not matter any connectivity between basins, rather we care about the probability mass in each basin and can then perform discrete sampling from the vector of probability masses to determine which basin to jump to. No graphs.

A quite different computation and analysis utilizes the *locality* in data space of objects that are localised in physical space for which we want to explore all configurations in representation space. We call such a collection of data/objects/representation a *basin*, borrowing the term from quantum chemistry where the term refers to a local minima in energy hence local maxima in probability. We might also call it a *resolution cell* in model space around state x , since it is all configurations that the GPR instrument cannot reliably distinguish from x – or one component of state x that contributes to the likelihood at one localised region in data space.

The term ‘resolution’ refers to the resolution of the equipment – spatial and other – that sets a kind of ambiguity in representation that is inherent in the observation process.

So a suitable data structure for a basin could be an equivalence class of interpretations of the data (at some data locality):

- representative model representation
- list of equivalent representations (each with different numbers of objects)
- local Gaussian approximation for each number (or other suitable density approximation)
- probability mass
- local sampler

We might be able to discover a prototype for the structure of a basin by offline computation, purely based on the forward map and noise statistics. What set of objects in some physical location produce effectively the same data in some region of data space? Maybe formalize that as: what subset of representation space

$$\mathcal{X} = \bigcup_{k=0}^{\infty} \mathcal{X}_k,$$

where $\mathcal{X}_k$ is the space of k objects, produces that same data projected onto a local-region of data space, in the sense that the likelihood is essentially the same? Suppose we can determine a region in data space $D_{\text{loc}}(x, y)$ that defines a kind of local-region in data space centred on data location (x, y) . Then, for a representative model $\mathbf{x}$, that produces data in that region D_{loc} we define the set

$$X_{\text{res}}(\mathbf{x}) = \{\mathbf{x}' : |L(\mathbf{x}') - L(\mathbf{x})| < \varepsilon\},$$

where data is projected onto $D_{\text{loc}}(x, y)$ for the comparison of likelihoods. So we have a notion of locality in data space and a resolution in representation space.

Given such a set of basins, we might be able to determine sensible density approximations, samplers, etc, that estimate probability mass.

Once data is measured, we run an MCMC, but not as previously known. The MCMC has the task of determining the data-dependent basins, moving amongst meta representations, and perhaps determining the probability mass in each basin, perhaps adaptively. Given a dictionary that contains basins, masses, moving around them is a task in sampling from a discrete distribution, which is jolly easy. So the MCMC is used to explore, only, which builds up a data structure of possible interpretations of the data, that is then sampled using simple, efficient discrete samplers.

A possibly relevant paper is:

- **Mapping Energy Landscapes of Non-Convex Learning Problems** Maria Pavlovskaja, Kewei Tu, Song-Chun Zhu, <https://arxiv.org/abs/1410.0576>

And a thought: We might generalise the proposal mechanism (generalizing the birth move) using the retrospective MCMC idea in:

Retrospective Markov chain Monte Carlo methods for Dirichlet process hierarchical models, Omiros Papaspiliopoulos and Gareth O. Roberts *Biometrika* (2008), 95, 1, pp. 169–186